a
True for each sample
True for the distribution
C1 Local linear
C2 Local nonlinear
C3 Symmetry
C4 Distributional statistic
P
Jensen gap
g
0
dE/dt
Ax = 0
∇·u = 0, ∇·B = 0
Homogeneous linear boundary conditions
c(x) = 0
Stationary PDE residual
g#μ = μ
B ↦ −B
Rotation Rπ (VKS)
𝔼μ[ℒφ] = 0
Energy balance
𝔼[ε] = 𝔼[P]
Matched mechanism
Potentials B = ∇×A
or
Leray projection
Nonlinear projection
or
Penalty, guidance
Equivariant vθ
and
Invariant base distribution
Stationarity loss
or
Exponential tilting
b
Energy balance (C4)
Illustrative shapes
True distribution
Per-sample projection
Distributional correction
μ
mean
0
dE/dt
Dirac at 0
qθ
Samples
Weights wi
𝔼 = 0, Var > 0
𝔼 = 0, Var = 0
𝔼 = 0, Var > 0